

```

#include <stdio.h>
#include <stdlib.h>
#define TABLE_SIZE 3
struct node {
int data;
struct node *next;
};
struct node *head[TABLE_SIZE] = {NULL};
void insert(int val) {
int i = val % TABLE_SIZE;
struct node *newnode = (struct node *)malloc(sizeof(struct node));
if (newnode == NULL) {
perror("Memory allocation failed");
exit(EXIT_FAILURE);
}
newnode->data = val;
newnode->next = NULL;
if (head[i] == NULL) {
head[i] = newnode;
} else {
struct node *c = head[i];
while (c->next != NULL) {
c = c->next;
}
c->next = newnode;
}
}
void display() {
for (int i = 0; i < TABLE_SIZE; i++) {
printf("Entries at index %d: ", i);
struct node *temp = head[i];
if (temp == NULL) {
printf("No Hash Entry\n");
} else {
while (temp != NULL) {
printf("%d -> ", temp->data);
temp = temp->next;
}
printf("NULL\n");
}
}
}
int main() {
int opt, val;

```

```
while (1) {  
    printf("\nPress 1. Insert\t 2. Display \t3. Exit \n");  
    scanf("%d", &opt);  
    switch (opt) {  
        case 1:  
            printf("\nEnter a value to insert into hash table: ");  
            scanf("%d", &val);  
            insert(val);  
            break;  
        case 2:  
            display();  
            break;  
        case 3:  
            exit(0);  
        default:  
            printf("\nInvalid option\n");  
    }  
}  
return 0;  
}
```